Failure of Unconditional Extrapolation for Operator-Valued Dyadic Paraproducts

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Status. Candidate full counterexample, likely valid, for expert review.

1 Source question

Let $(\mathcal{F}_j)_{j\geq 0}$ be the usual dyadic filtration on the circle and write E_j and $d_j = E_j - E_{j-1}$ for conditional expectation and martingale difference. For a von Neumann algebra $\mathcal{M}$ with a normal semifinite faithful trace, Mei defines the left dyadic paraproduct

$$\pi_\varphi f = \sum_j (d_j\varphi)(E_{j-1}f) \quad \text{on } L^p(\mathbb{T}; L^p(\mathcal{M})).$$

Theorem 1.2 of [1] proves that boundedness for one $1 < p < \infty$ extrapolates to all such p provided $\varphi \in \mathrm{BMO}_{\mathcal{M}}(\mathbb{T}, \mathcal{M})$. The paper then asks whether that operator-norm BMO assumption can be removed [1, Open Question, PDF p. 9].

> As mentioned before, when $\mathcal{M} = \mathbb{C}$, the condition $\varphi \in \mathrm{BMO}_{\mathcal{M}}(\mathbb{T}, \mathcal{M})$ in Theorem 1.2 is not necessary since we have
>
> $$||\varphi||_{\mathrm{BMO}_{\mathcal{M}}(\mathbb{T},\mathcal{M})} \leq c||\pi_\varphi||_{L^p\to L^p}, \tag{3.16}$$
>
> for any p. But (3.16) does not hold for general von Neumann algebra $\mathcal{M}$ unless we replace $||\varphi||_{\mathrm{BMO}_{\mathcal{M}}(\mathbb{T},\mathcal{M})}$ by a smaller norm $||\varphi||_{\mathcal{BMO}_c(\mathbb{T},\mathcal{M})}$.
>
> **Open Question.** Can we remove the assumption $\varphi \in \mathrm{BMO}_{\mathcal{M}}(\mathbb{T}, \mathcal{M})$ in Theorem 1.2?

Equivalently, the question asks whether boundedness of π_φ on one $L^p(\mathbb{T}; L^p(\mathcal{M}))$ alone forces boundedness on every such space.

2 Claimed contribution

Theorem 1. *There are a semifinite tracial von Neumann algebra $\mathcal{M}$ and an $\mathcal{M}$-valued Bochner integrable symbol φ such that*

$$\|\pi_\varphi\|_{L^2(\mathbb{T};L^2(\mathcal{M}))\to L^2(\mathbb{T};L^2(\mathcal{M}))} \leq 1,$$

but, for every $2 < p < \infty$,

$$\pi_\varphi : L^p(\mathbb{T}; L^p(\mathcal{M})) \longrightarrow L^p(\mathbb{T}; L^p(\mathcal{M})) \quad \textit{is unbounded.}$$

The symbol belongs to $L^q(\mathbb{T}; \mathcal{M})$ in operator norm for every finite q, but $\|\varphi\|_{\mathrm{BMO}_{\mathcal{M}}} = \infty$. Hence the open question has a negative answer.

3 Proof intuition

The one-sided paraproduct can see columns and rows very differently. In an $n \times n$ matrix corner, put successive martingale differences in the first row: $d_k\varphi = r_k e_{1k}$. On L^2, martingale orthogonality and the identity $\|e_{1k}x\|_2 = \|e_{kk}x\|_2$ turn the output norm into a sum of mutually orthogonal input rows, so the norm is at most one. On the identity matrix, however, the output is the row $(r_1, \ldots, r_n)$, whose only singular value is $\sqrt{n}$. Its ratio to the Schatten-p norm of the identity is $n^{1/2-1/p}$.

A single finite block gives only a large norm, not an unbounded operator. We place the nth block on its own exponentially short dyadic interval and use mutually orthogonal matrix corners. The resulting infinite symbol is Bochner integrable, retains the uniform L^2 contraction, and contains all the divergent L^p tests simultaneously.

4 Construction

Identify $\mathbb{T}$ with $[0, 1)$ and let

$$\mathcal{H} = \bigoplus_{n\geq 1} \mathbb{C}^n, \qquad \mathcal{M} = B(\mathcal{H}),$$

equipped with its usual trace. In the nth summand let $e^{(n)}_{ab}$ denote the standard matrix units and let $P_n = \sum_{k=1}^n e^{(n)}_{kk}$.

For $n \geq 1$, set

$$I_n = [2^{-n}, 3 \cdot 2^{-(n+1)}), \qquad m_n = n + 1.$$

Thus I_n is an atom of $\mathcal{F}_{m_n}$, has length $2^{-(n+1)}$, and the intervals I_n are pairwise disjoint. On I_n let $r_{n,1}, \ldots, r_{n,n}$ be the local Rademacher functions: after extension by zero outside I_n, they satisfy

$$r_{n,k} \text{ is } \mathcal{F}_{m_n+k}\text{-measurable}, \qquad E_{m_n+k-1} r_{n,k} = 0, \qquad |r_{n,k}| = 1_{I_n}.$$

Define

$$\varphi(t) = \sum_{n=1}^{\infty} \sum_{k=1}^{n} r_{n,k}(t) e^{(n)}_{1k}. \tag{1}$$

At any t, at most one outer summand is nonzero. On I_n the operator $\varphi(t)$ is a rank-one row of length $\sqrt{n}$, so

$$\|\varphi(t)\|_{\mathcal{M}} = \sqrt{n} \quad (t \in I_n).$$

Consequently

$$\int_{\mathbb{T}} \|\varphi(t)\|^q_{\mathcal{M}}\, dt = \sum_{n\geq 1} 2^{-(n+1)} n^{q/2} < \infty \qquad (1 \leq q < \infty).$$

In particular, (1) is a legitimate Bochner integrable $\mathcal{M}$-valued symbol.

The placement in the filtration gives the exact martingale differences

$$d_j\varphi = \sum_{\substack{n\geq 1,\ 1\leq k\leq n \\ m_n+k=j}} r_{n,k} e_{1k}^{(n)}. \tag{2}$$

The sum in (2) is finite for each j. Indeed, coarser local Rademachers are already $\mathcal{F}_{j-1}$-measurable, while finer ones have zero conditional expectation at both levels $j-1$ and j.

5 The uniform L^2 bound

Lemma 1. *The series defining π_φ converges on $L^2(\mathbb{T}; S^2(\mathcal{H}))$ and $\|\pi_\varphi\|_{2\to 2} \leq 1$.*

Proof. Start with a finite martingale f. Terms belonging to different global martingale levels are orthogonal in L^2. At a common level, terms from different n have orthogonal left ranges. Hence (2) gives

$$\|\pi_\varphi f\|_2^2 = \sum_{n\geq 1}\sum_{k=1}^{n} \left\| r_{n,k} e_{1k}^{(n)} E_{m_n+k-1} f \right\|_2^2. \tag{1}$$

All norms here include integration in t and the Hilbert–Schmidt norm. Because $I_n \in \mathcal{F}_{m_n}$ and $m_n + k - 1 \geq m_n$,

$$1_{I_n} E_{m_n+k-1} f = E_{m_n+k-1}(1_{I_n} f).$$

For every Hilbert–Schmidt operator x,

$$\|e_{1k}^{(n)} x\|_2 = \|e_{kk}^{(n)} x\|_2.$$

Conditional expectation is an L^2 contraction and commutes with the constant matrix unit, so each summand in (1) is at most

$$\|1_{I_n} e_{kk}^{(n)} f\|_2^2.$$

The intervals I_n are disjoint and all the projections $e_{kk}^{(n)}$ are mutually orthogonal. Therefore

$$\|\pi_\varphi f\|_2^2 \leq \sum_{n\geq 1}\sum_{k=1}^{n} \|1_{I_n} e_{kk}^{(n)} f\|_2^2 \leq \|f\|_2^2.$$

The same estimate for partial sums makes the full orthogonal series convergent. Density then gives the asserted contraction on all of L^2. □

6 Failure on every L^p above two

Fix n and use the $\mathcal{F}_{m_n}$-measurable test function

$$f^{(n)}(t) = 1_{I_n}(t) P_n.$$

For every $1 \le k \le n$, $E_{m_n+k-1} f^{(n)} = f^{(n)}$. Thus

$$\pi_\varphi f^{(n)} = \sum_{k=1}^{n} r_{n,k} e_{1k}^{(n)}. \tag{3}$$

The projection P_n has n singular values equal to one, whereas the row in (3) has exactly one nonzero singular value, namely $\sqrt{n}$. It follows that, for every finite p,

$$\|f^{(n)}\|_{L^p(\mathbb{T};S^p)} = |I_n|^{1/p} n^{1/p}, \qquad \|\pi_\varphi f^{(n)}\|_{L^p(\mathbb{T};S^p)} = |I_n|^{1/p}\sqrt{n}.$$

Hence

$$\frac{\|\pi_\varphi f^{(n)}\|_p}{\|f^{(n)}\|_p} = n^{1/2-1/p}. \tag{4}$$

For each fixed $p > 2$, the right-hand side tends to infinity. Therefore π_φ is unbounded on every $L^p(\mathbb{T}; S^p)$ with $p > 2$. Together with Lemma 1, this proves Theorem 1.

Finally, the construction lies exactly outside the hypothesis in the source. Since every local Rademacher has mean zero on I_n, one has $\varphi_{I_n} = 0$, and therefore

$$\left(\frac{1}{|I_n|} \int_{I_n} \|\varphi - \varphi_{I_n}\|_{\mathcal{M}}^2 \, dt \right)^{1/2} = \sqrt{n}.$$

Taking the supremum over dyadic intervals proves $\|\varphi\|_{\mathrm{BMO}_{\mathcal{M}}} = \infty$.

Remark 1 (Finite tracial variant)**.** *If a finite trace is preferred, replace $B(\mathcal{H})$ by $\prod_{n\ge1} M_n$ with $\tau((x_n)) = \sum_{n\ge1} 2^{-n}\,\mathrm{tr}_n(x_n)$, where tr_n is normalized. The same symbol and the same proof work blockwise. Both norms in the nth test acquire the same factor $2^{-n/p}$, so the ratio* (4) *is unchanged.*

7 Verification, limitations, and novelty status

The script `code/verify_counterexample.py` independently checks the local martingale-difference identities, random finite-block versions of the L^2 contraction for $n \le 8$, the rank-one singular-value computation, and the exact ratios (4) for several p. These computations are sanity checks; the proof above is exact and does not depend on them.

The construction answers only the yes/no extrapolation question. It does not characterize all symbols whose paraproducts extrapolate, nor does it settle the still broader problem of characterizing L^2 boundedness of operator-valued paraproducts.

A bounded novelty search was performed on 11 August 2026: the run's title, abstract, source-signal, result, and attempt indexes were searched for the arXiv id and the core phrases; exact and keyword searches were made for an operator-valued paraproduct bounded on L^2 but unbounded on L^p; and later arXiv papers citing the source were inspected. In particular, Wei–Zhang still state that general L^2 boundedness characterization is open and formulate extrapolation with the same strong BMO hypothesis [2]. No matching counterexample was located. This is bounded negative evidence, not a priority claim.

Human-review recommendation. Send to a specialist in noncommutative martingales. The critical audit points are the global martingale-difference identity (2), the orthogonality in Lemma 1, and whether the source question is read in its literal ambient class of Bochner integrable $\mathcal{M}$-valued symbols.

References

[1] T. Mei, *An extrapolation of operator valued dyadic paraproducts*, J. London Math. Soc. (2) 81 (2010), no. 3, 650–662; arXiv:0709.4229.

[2] Z. Wei and H. Zhang, *Boundedness of operator-valued commutators involving martingale paraproducts*, arXiv:2401.08729.